



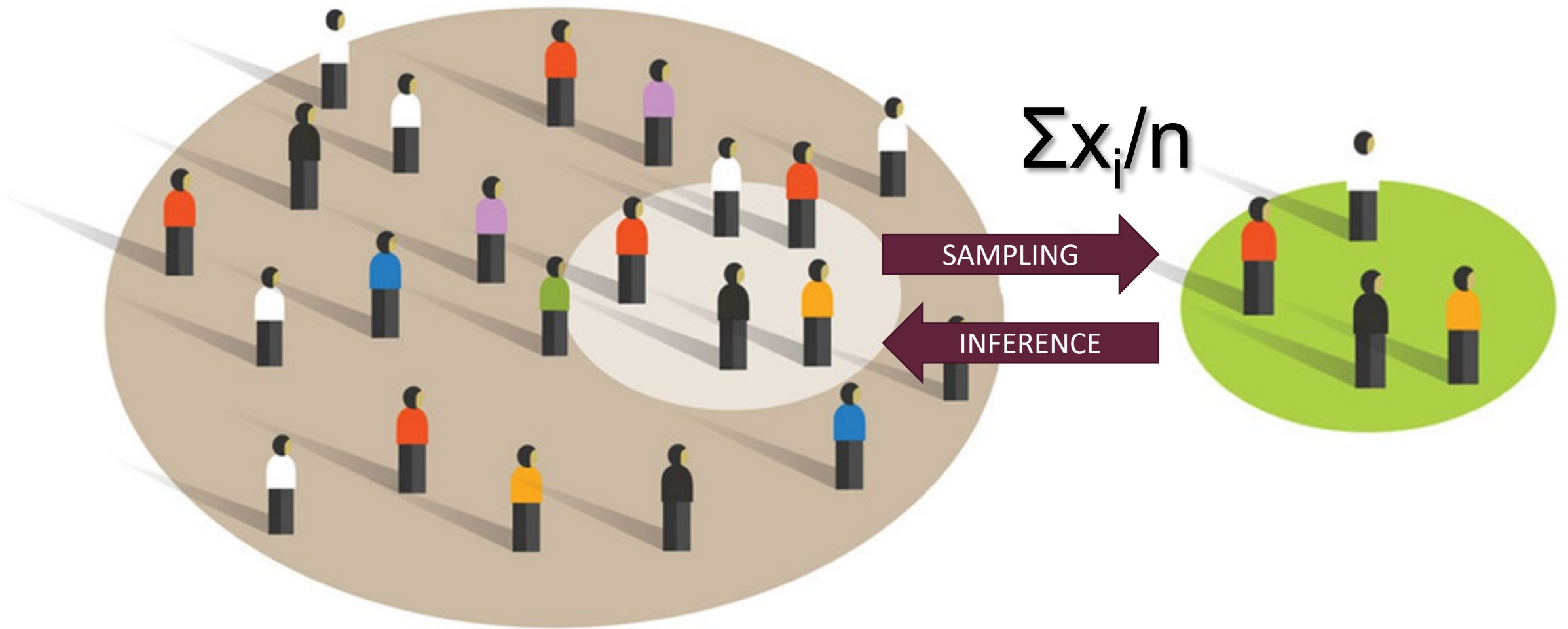
Survey Methods

2.1 - Survey error

- The Total Survey Error framework
- Understanding sources of error
- Cost, precision and bias: a trade-off
- Beyond error

Example: estimating mean blood pressure in a (large) population





SAMPLING



representation
error

ERROR

measurement
error

processing
error

inferential error

115.6 (110.1; 122.4)



MEASUREMENT

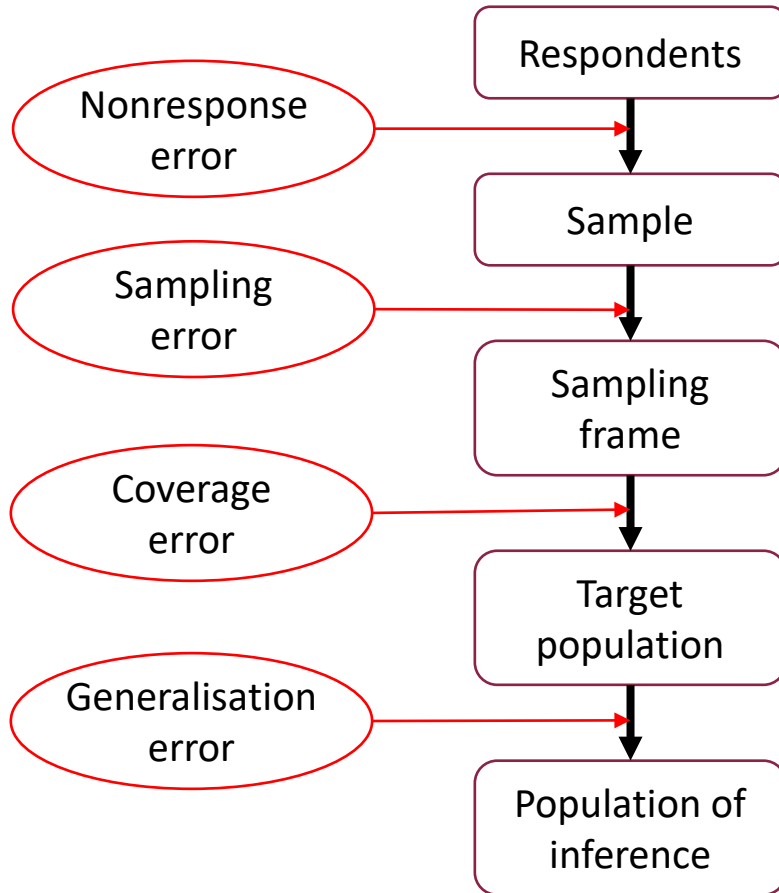


DATA MANAGEMENT



DATA ANALYSIS

Representation Error



Measurement And Processing Error

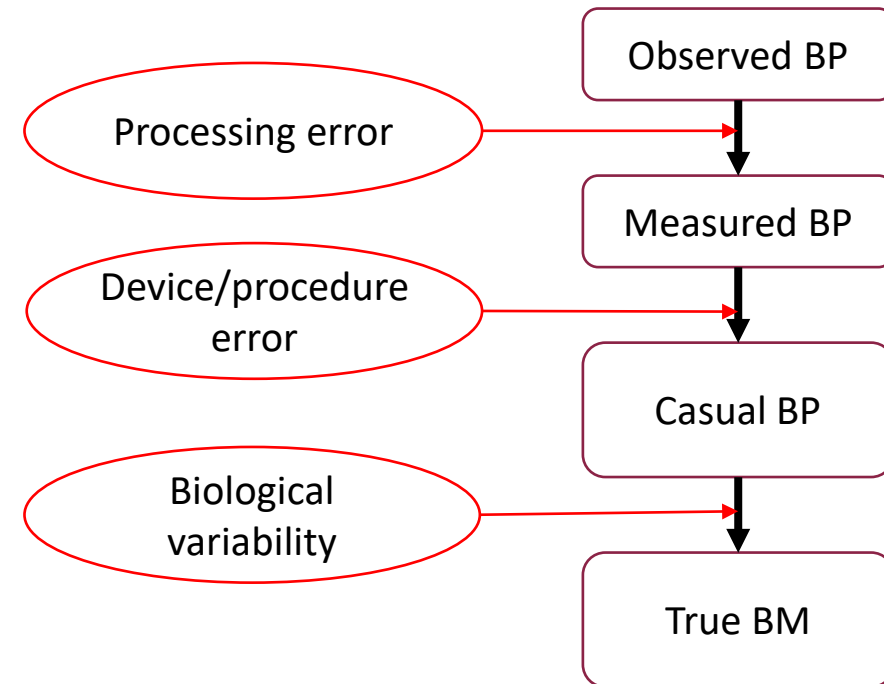
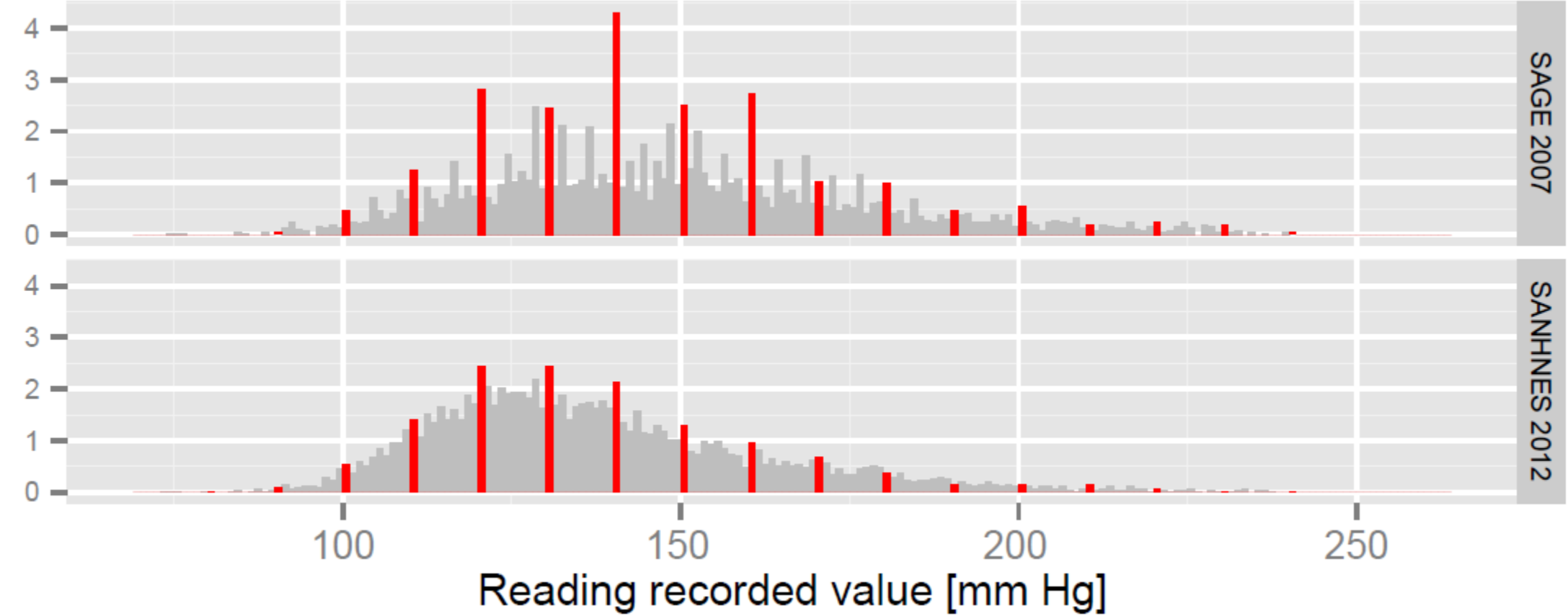
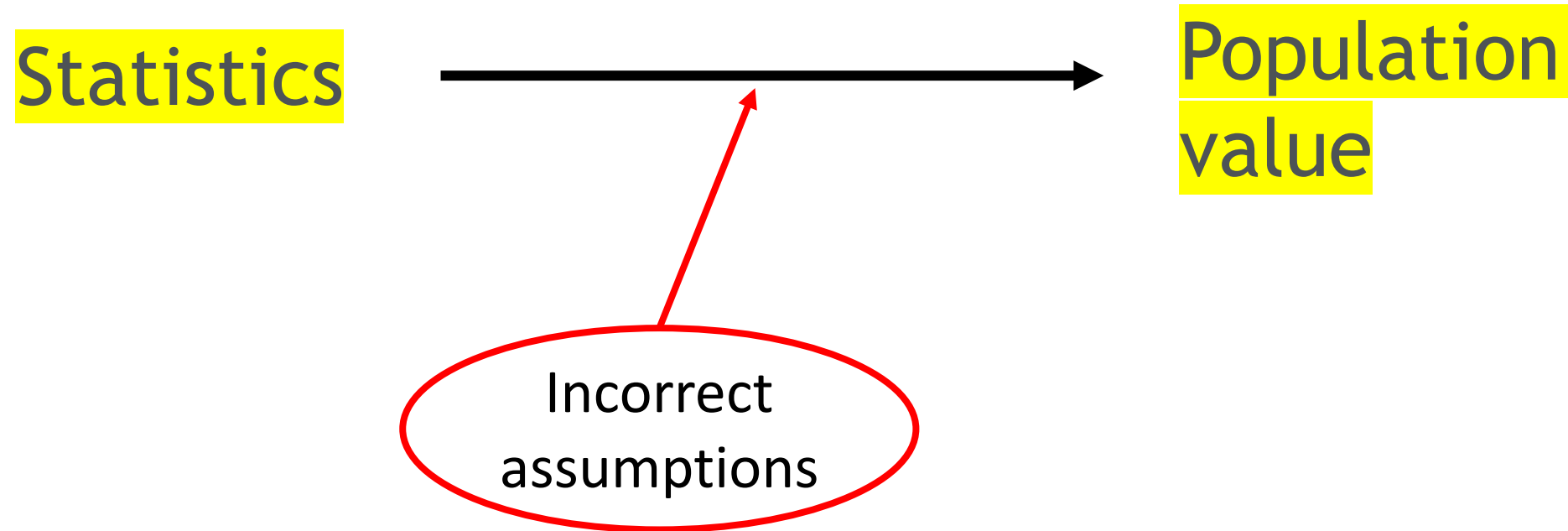


Figure 4.13: Distribution of first measurment of systolic blood pressure across surveys.

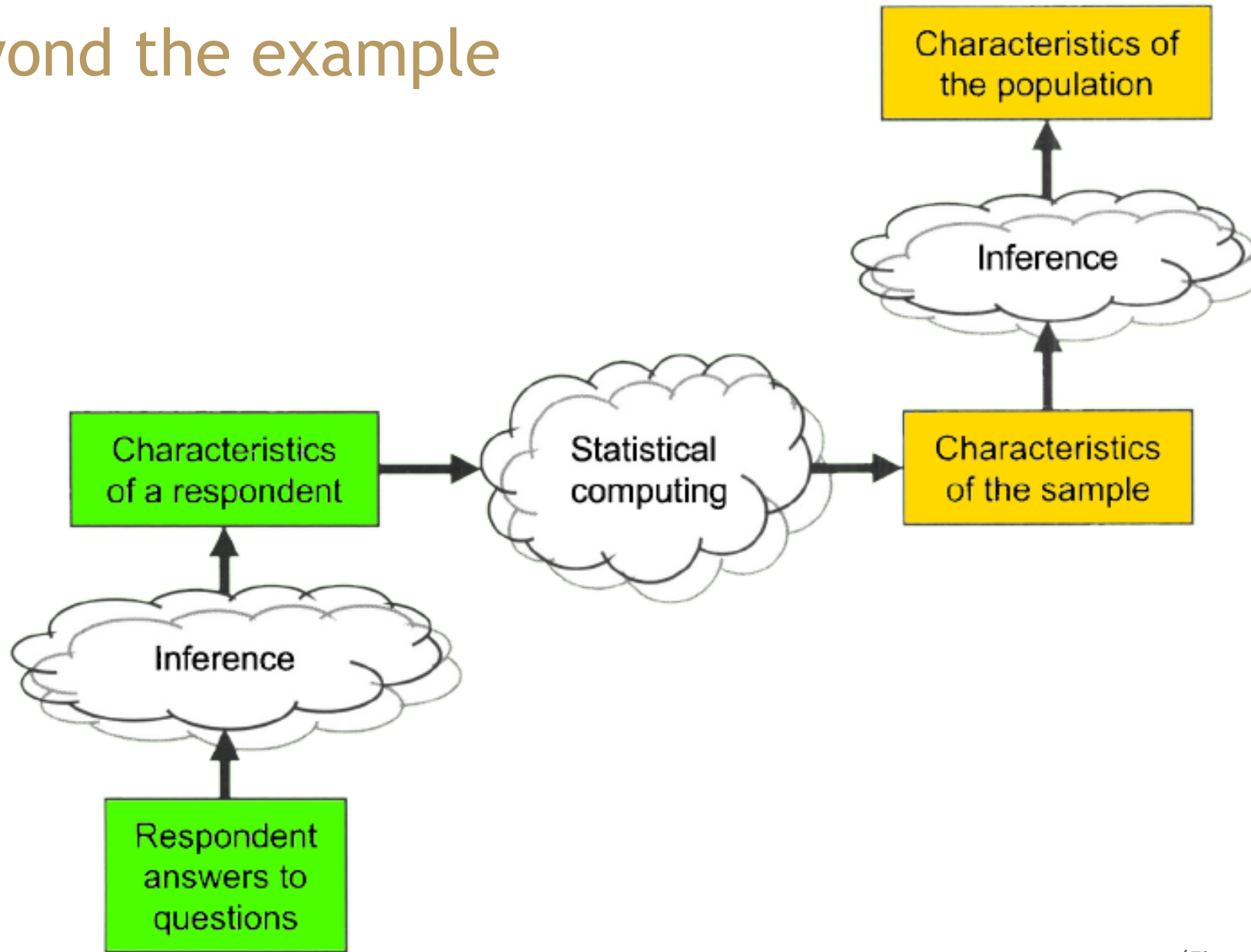


(Figure adapted from Cois, 2017:229)

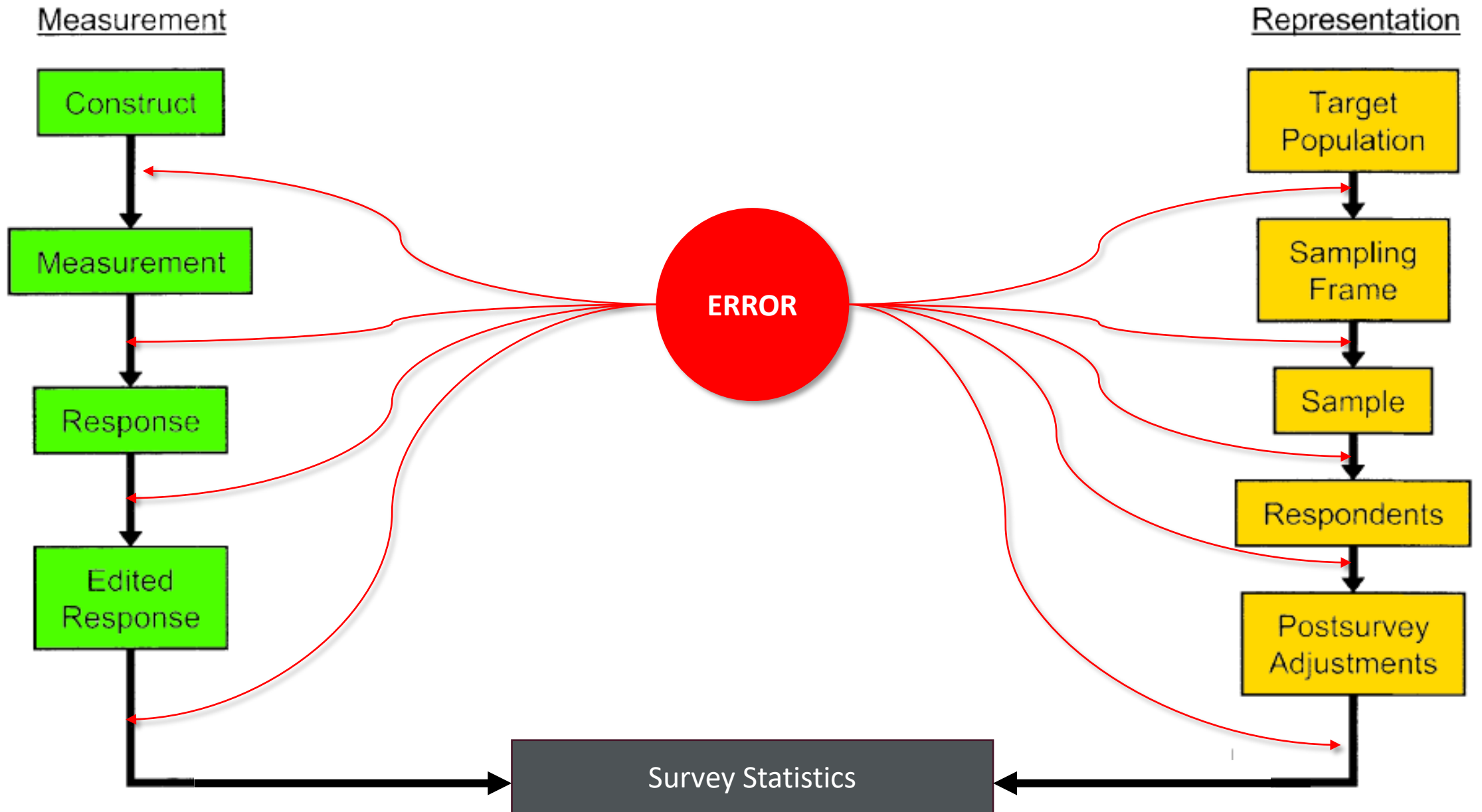
Estimation Error



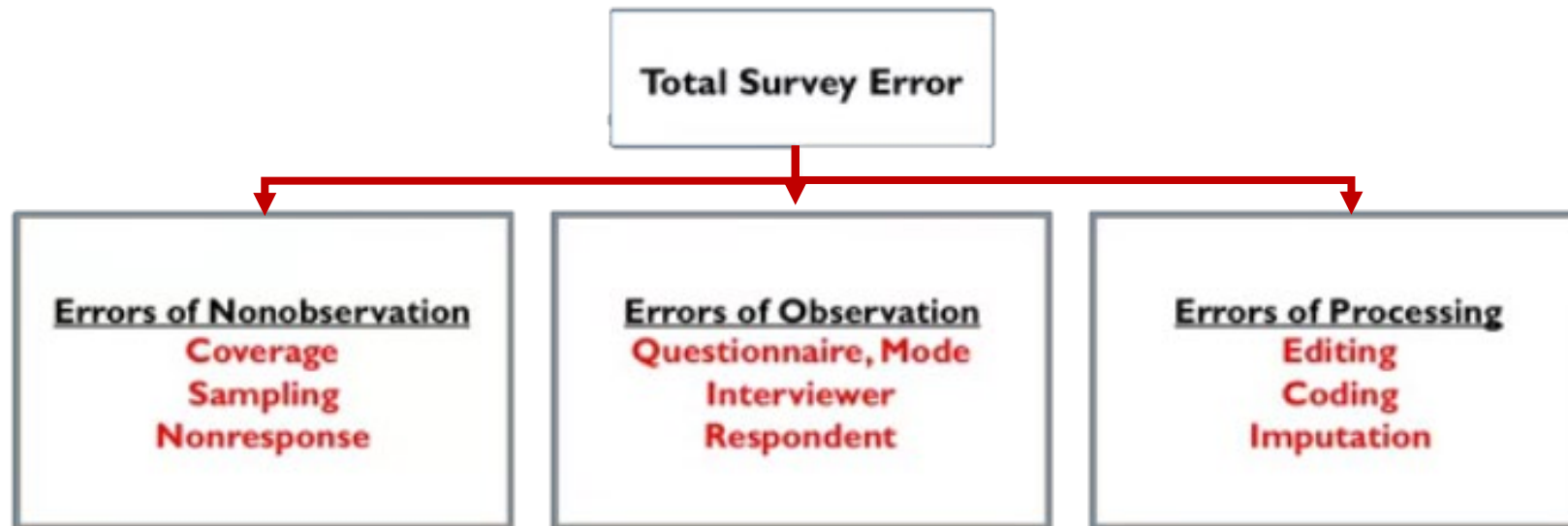
Beyond the example



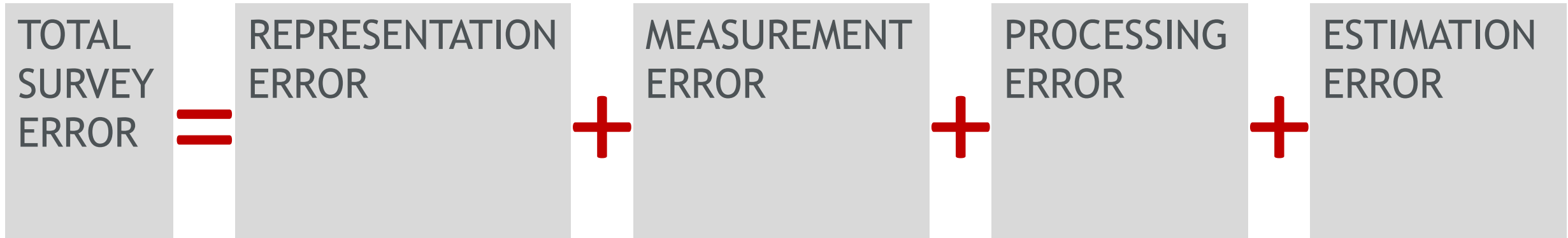
(Figure adapted from Groves et al. 2009:39)



(Figure adapted from Groves et al. 2009:42)



Total Survey Error (TSE) framework



MINIMISE **THIS**
WITHIN
CONSTRAINTS



Careful planning is required for allocating resources to the various stages of the survey process so that the major sources of error are controlled to acceptable levels. The goal is not to conduct every stage of the survey process as error-free as possible, because that would entail exceeding the survey budget and/or schedule by a considerable margin. Even under the best circumstances and given an unlimited budget and time, the potential for survey errors will always remain in some operations. Instead, the goal is to avoid the most egregious errors and control other errors to the extent that remaining errors are mostly inconsequential and tolerable.

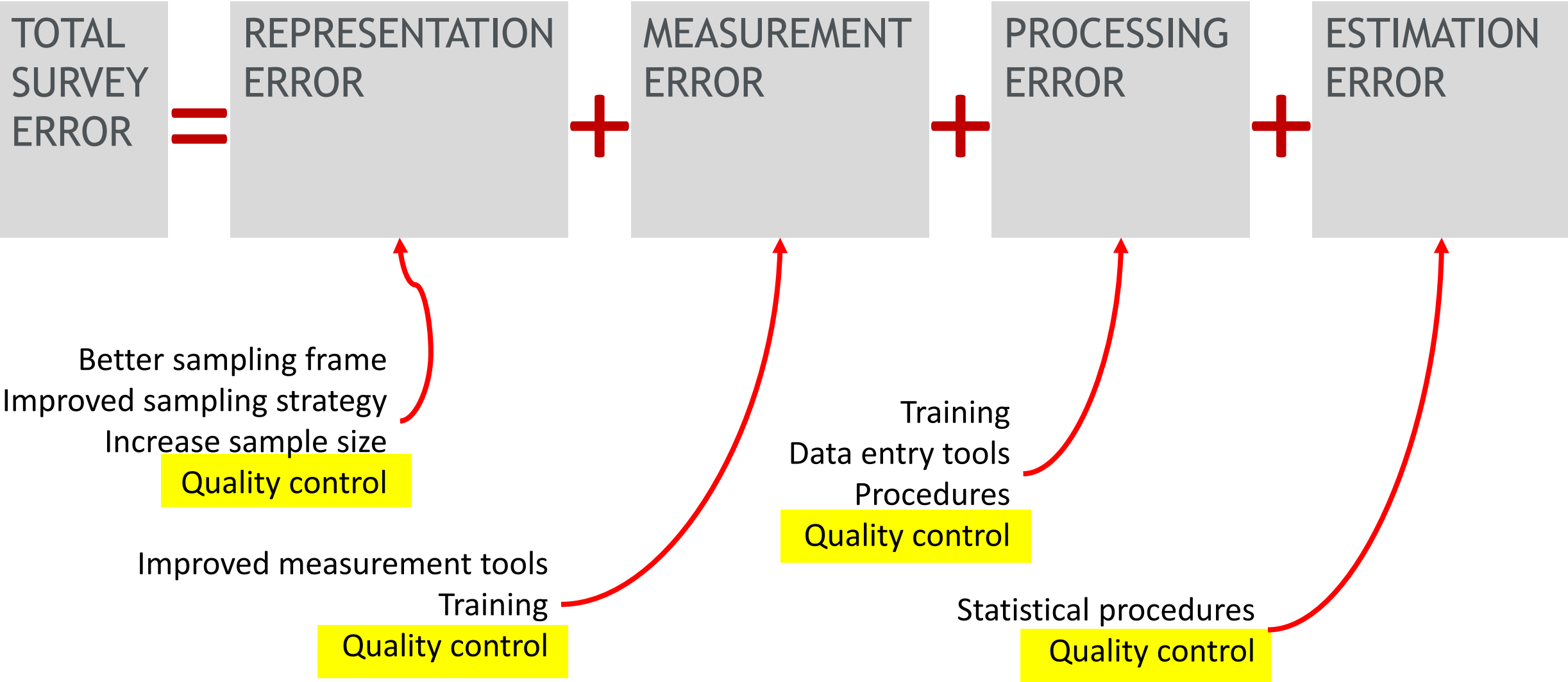
(Biemer 2010:821)

A 'balancing exercise'

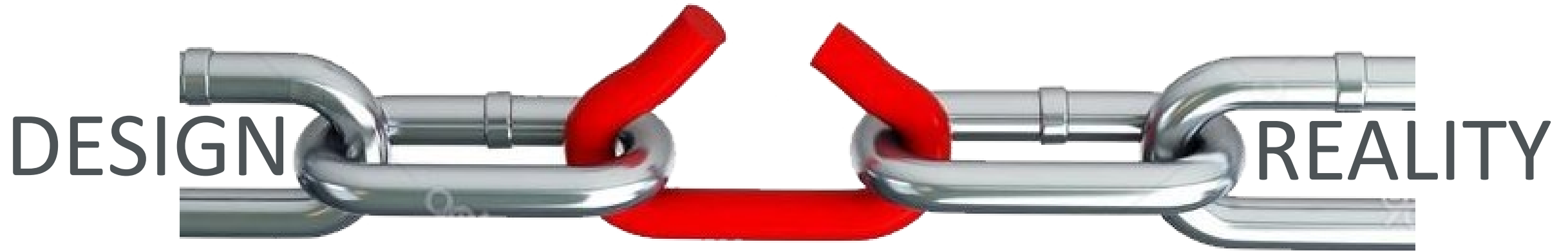
DISTRIBUTING
WISELY YOUR
RESOURCES



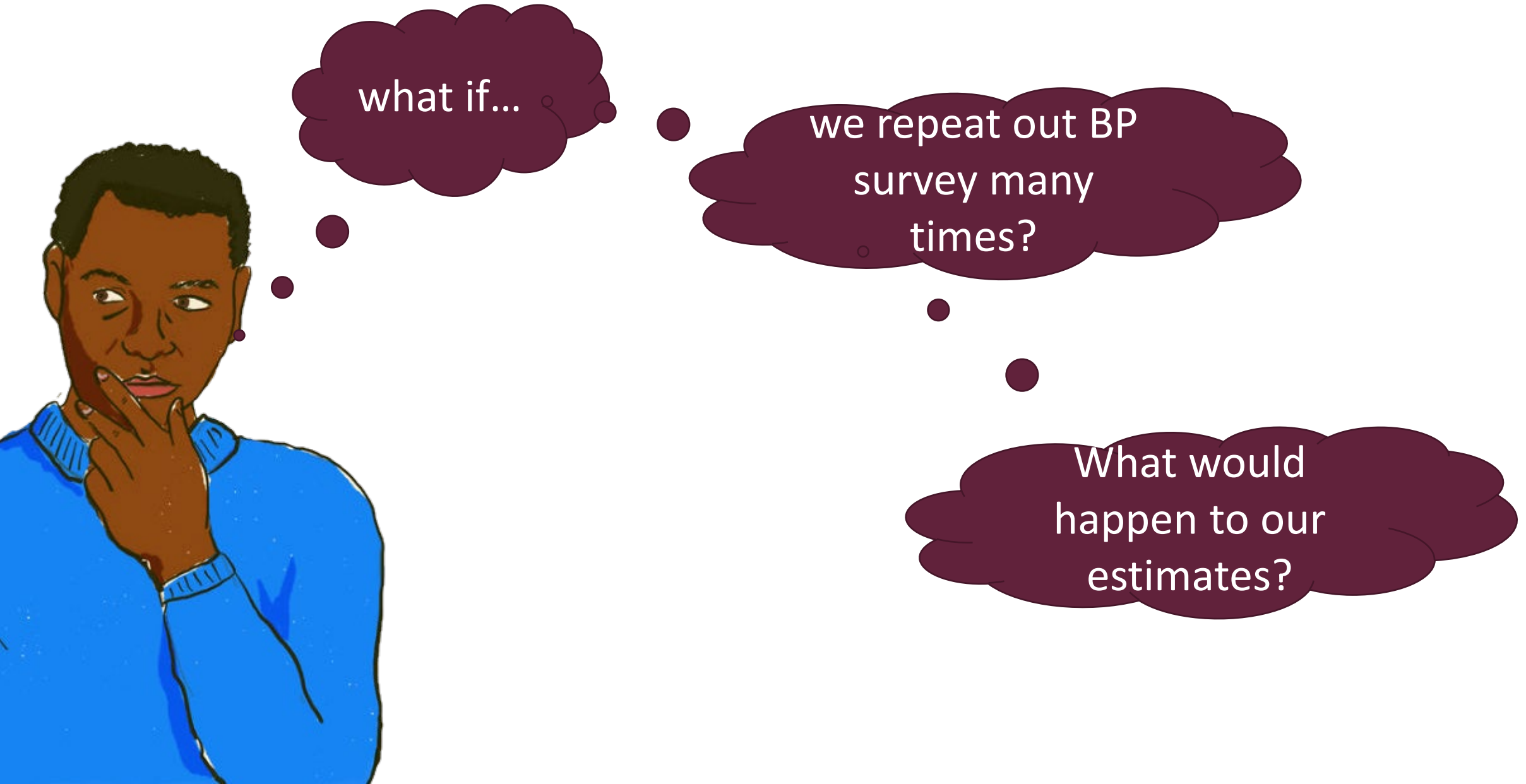
Total Survey Error (TSE) framework



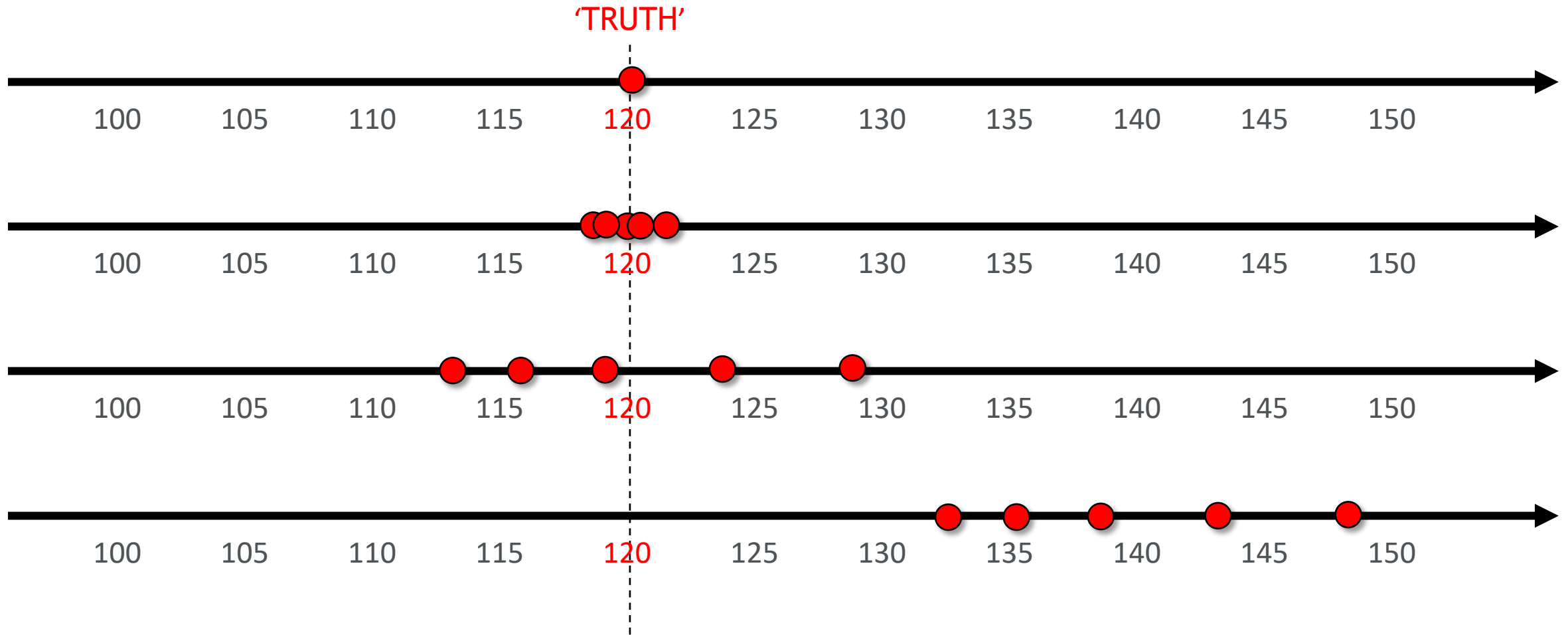
QUALITY CONTROL



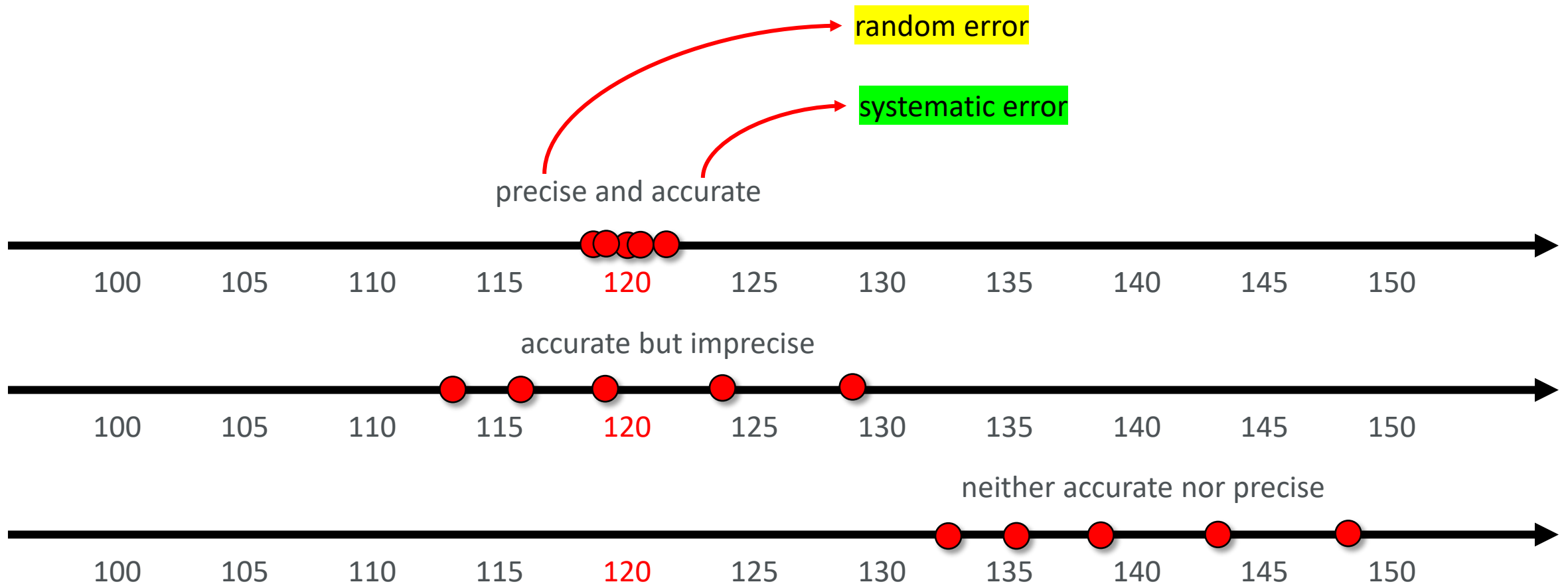
Systematic vs. Random Error



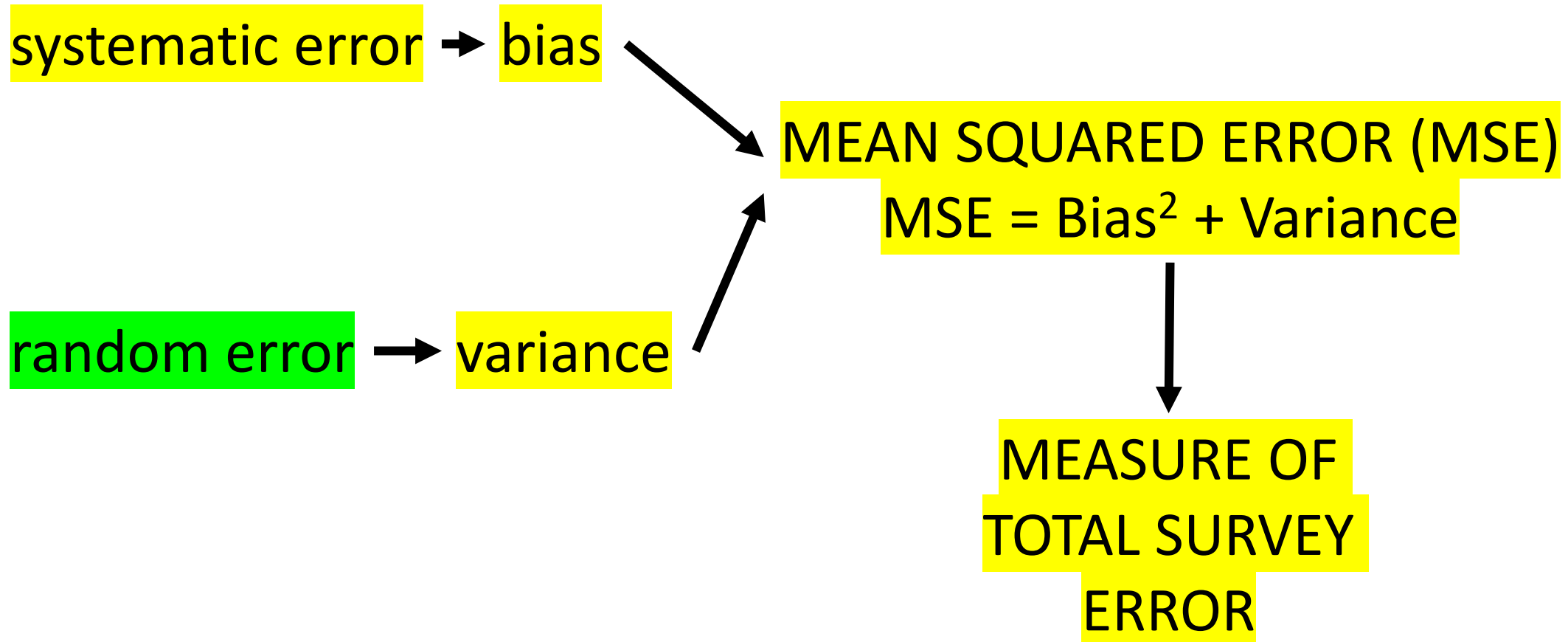
Systematic vs. Random Error



Systematic vs. Random Error

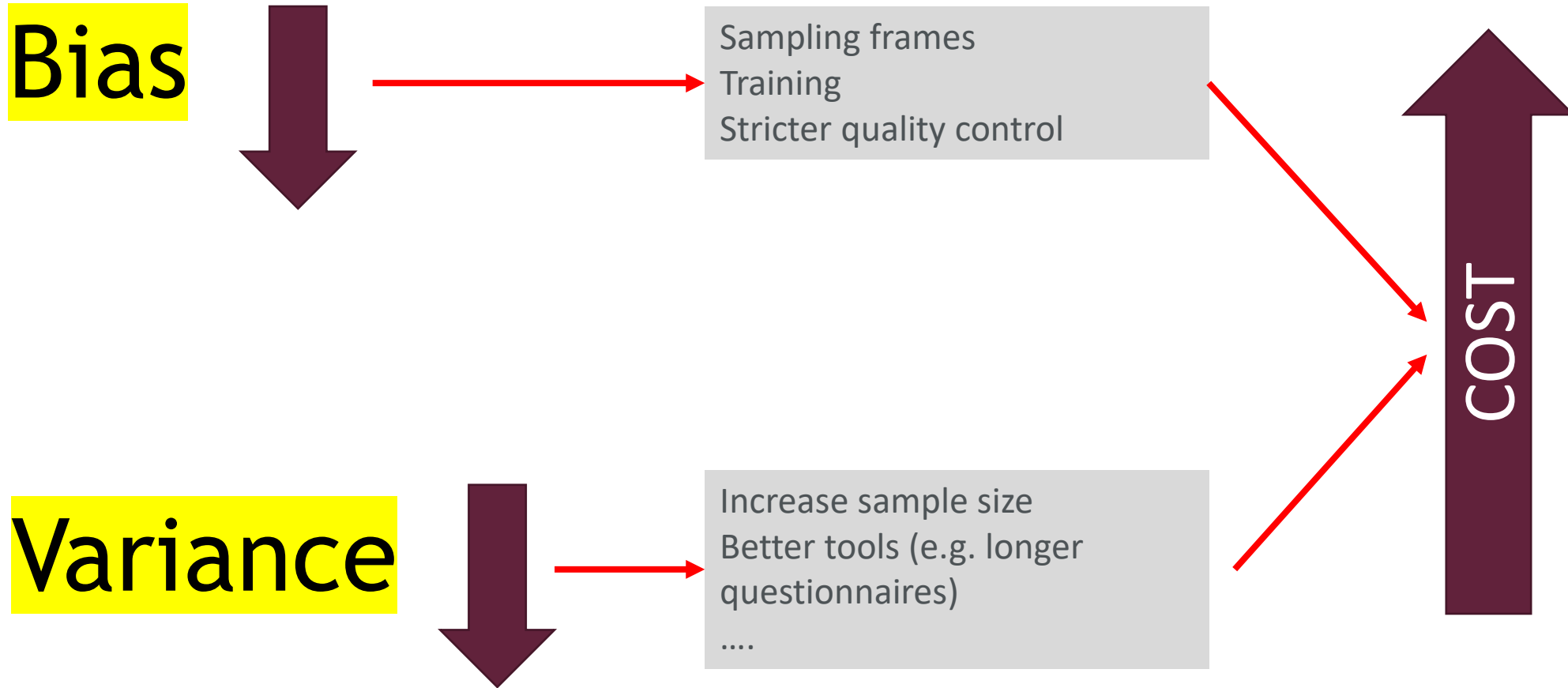


Systematic vs. Random Error



Quantifying MSE: see Biemer 2010

Cost, Precision and Bias



Examples:

Smartphone Participation in Web Surveys

(Peterson et al. 2017; Couper et al. 2017)

The problem: should we allow respondents to a **web survey** to use their **smartphones** to fill the survey questionnaire?



<7 inches



Responses from smartphones are on the rise...

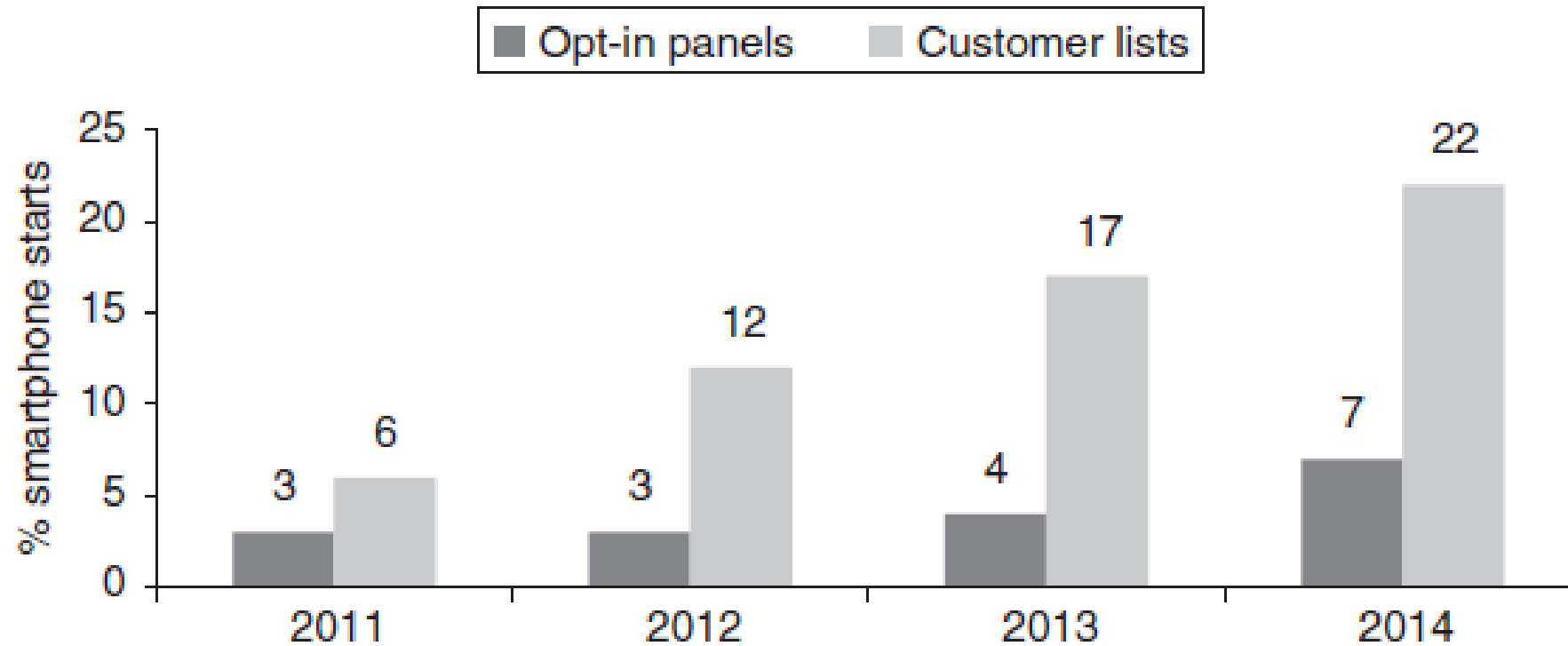


Figure 10.2 Prevalence of smartphone starts from the Decipher platform's highest volume accounts.

(Figure from Peterson et al. 2017:208)

(Potential) problems in allowing respondents to use smartphones

Long/complex survey instruments that are not suited to smartphones



Breakoffs

“...breakoff rates were 22% for desktop/laptop respondents, vs. 37.4% for mobile respondents.”

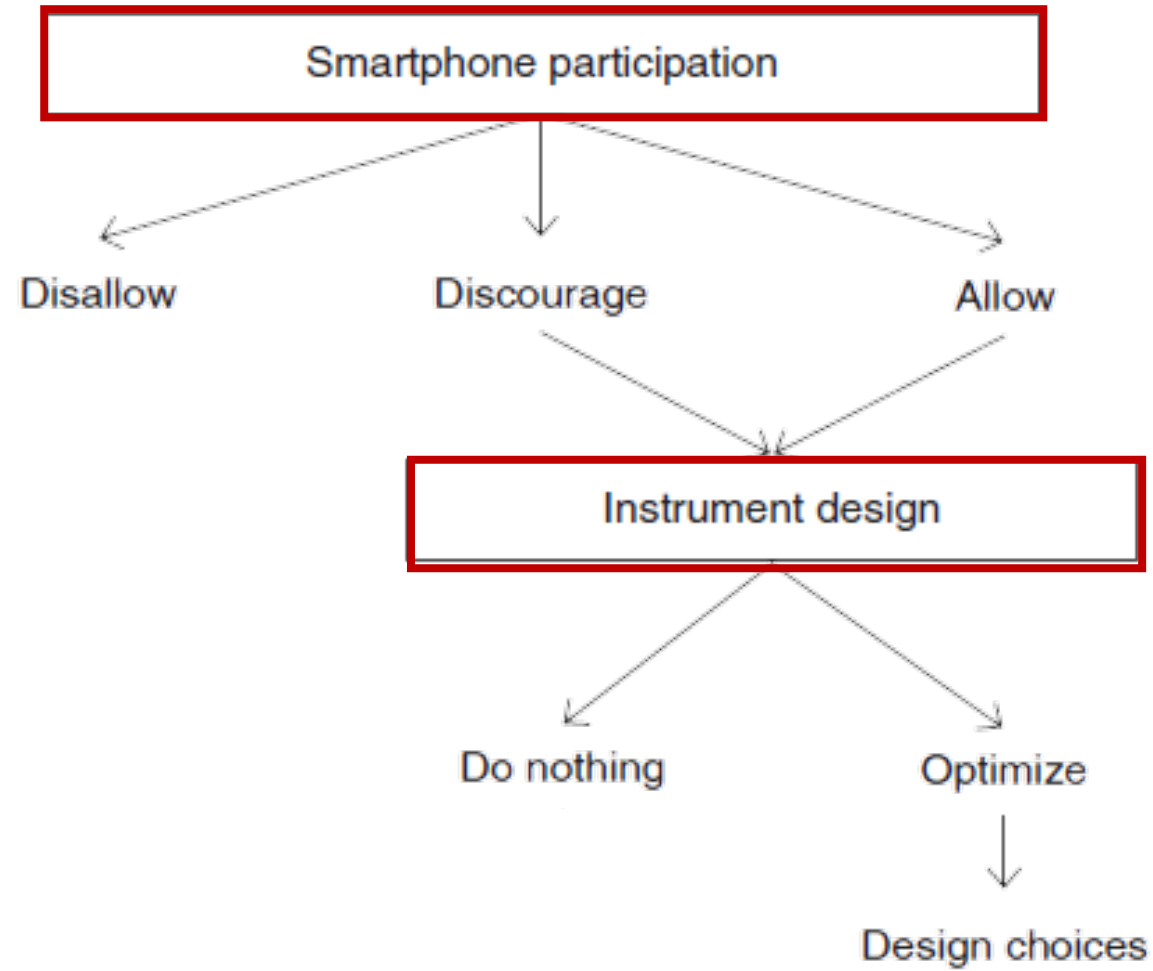
(Callegaro 2010)

Standardisation of mode of administration

Differential Symptom Reporting by Mode of Administration of the Assessment: Automated Voice Response System versus a Live Telephone Interview

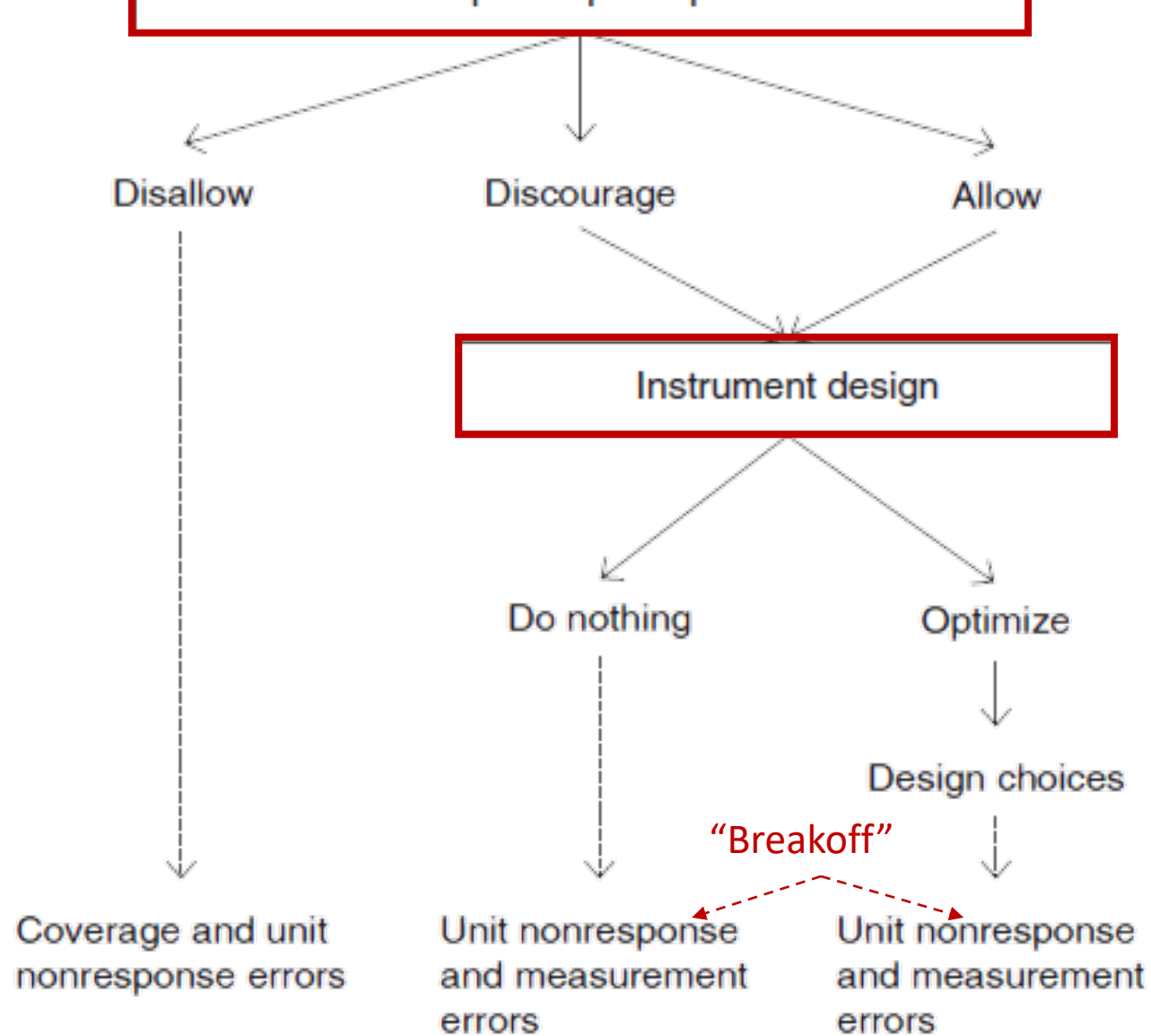
(Sikorskii et al. 2009)

CHOICES

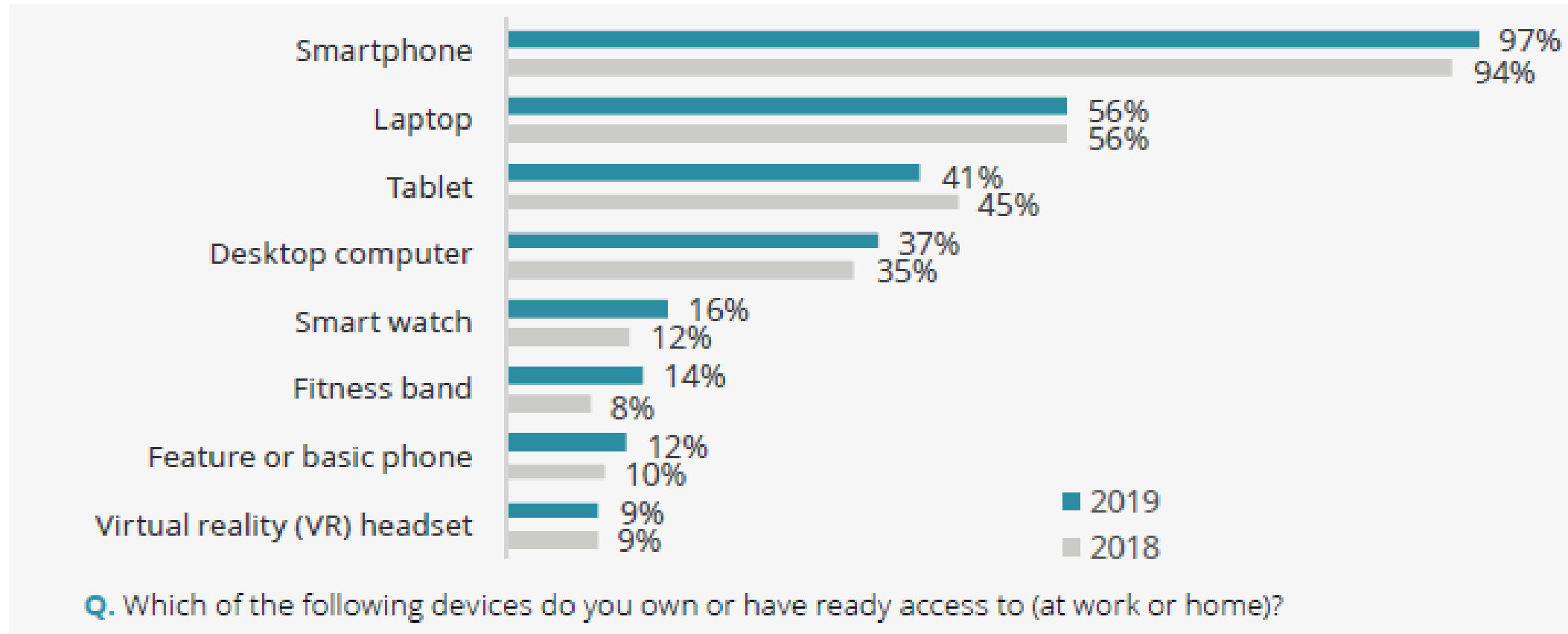


CHOICES

POTENTIAL ERROR

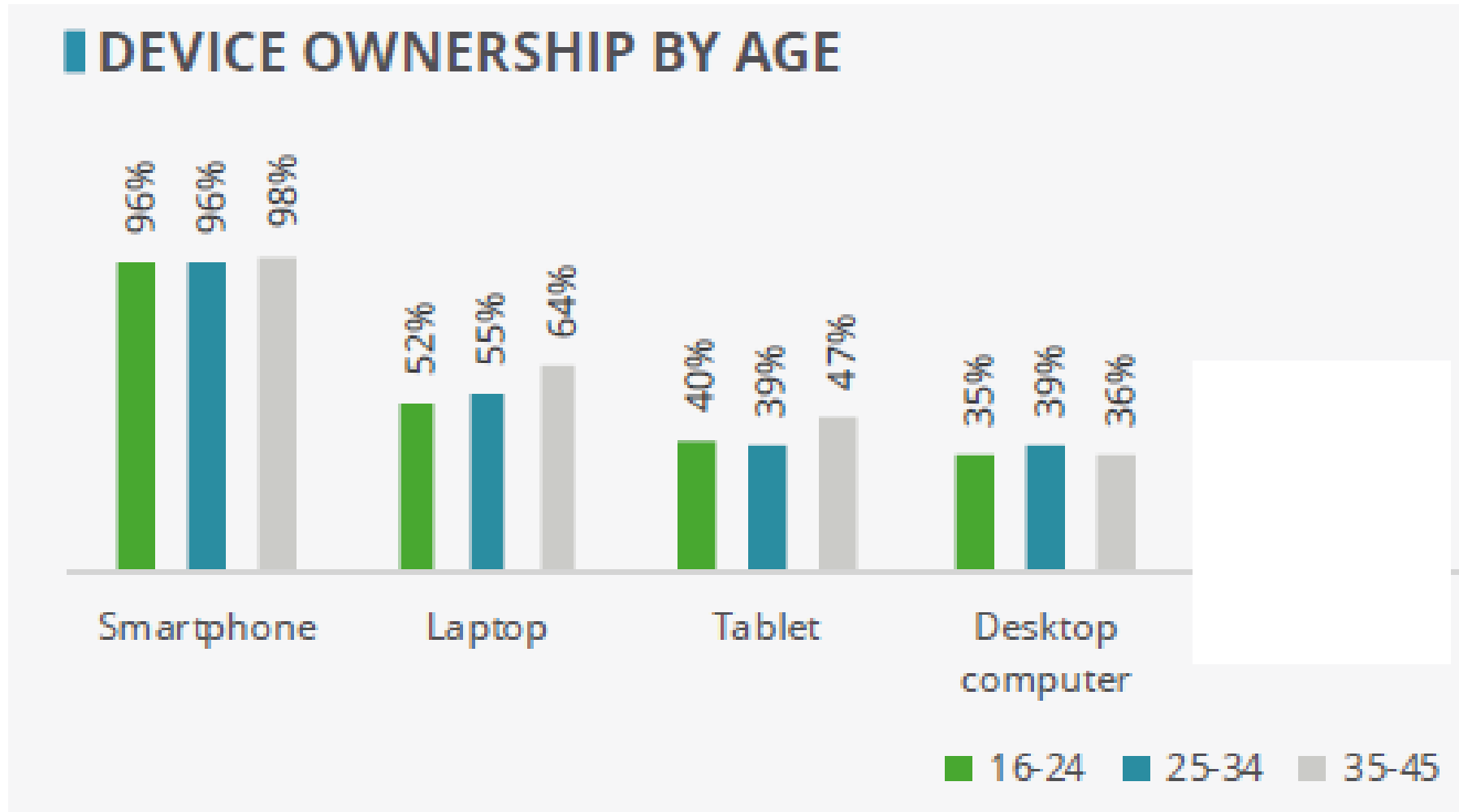


Risk: undercoverage



Source: Deloitte Touche Tohmatsu Limited. Global Mobile Consumer Survey 2019: South Africa Cut
<https://www2.deloitte.com/content/dam/Deloitte/za/Documents/technology-media-telecommunications/za-GMCS-2019-final-report.pdf>

Risk: differential undercoverage



Source: Deloitte Touche Tohmatsu Limited. Global Mobile Consumer Survey 2019: South Africa Cut
<https://www2.deloitte.com/content/dam/Deloitte/za/Documents/technology-media-telecommunications/za-GMCS-2019-final-report.pdf>

Risk: breakoff

Unit non-response

Table 7.2 Breakoff rates (%) for PC and mobile web users.

	PC web	Mobile web	Optimized for mobile?
Baker-Prewitt (2013) ^{a)}	5	18	No
		13	Yes
Barlas et al. (2015) ^{a)}	1.5	7.5	No
		5.5	Yes
Bosnjak et al. (2013)	6	12	No
Buskirk and Andrus (2014) ^{a)}	28.4	30.9	Yes
Couper and Peterson (2016)	13.4	27.7	No
Guidry (2012)	13	27	No
Horwitz (2014)	11.5	26.8	No
Mavletova (2013) ^{a)}			
Short survey	0.7	16.3	Yes
Long survey	2.3	20.3	Yes
Mavletova and Couper (2013) ^{a)}	2.9	13.6	Yes
Maxl (2014)	38	44	No
McGeeney and Marlar (2013)	0.3	1.3	No
		1.0	Yes
Stapleton (2013)	3.3	15.9	No
		6.3–10.2	Yes
Wells et al. (2013a) ^{a)}	1	4	Yes
Average (and 95% CI) from meta-analysis	5.5 (2.5; 9.6)	13.4 (8.4; 19.5)	

a) Random assignment to device; all others self-selected device.

(Table from Couper et al. 2017:140)

Risk: measurement error

Table 7.3 Measurement error in mobile web surveys: main findings and possible directions for further research.

	Main findings	Possible directions for further research
Grouping of questions		
Question-order effects	Similar to other survey modes, the context effect was found in mobile web surveys (Peytchev and Hill, 2010 ^a).	Comparison of context effects between mobile and PC web surveys.
Grouping of items on a page	No effect of the number of items presented on a page: no effect on response-order effects, the rate of nonsubstantive responses, and the length of answers to open-ended questions in mobile web surveys (de Bruijne and Wijnant, 2014b ^a ; Mavletova and Couper, 2016a ^a). A large number of questions presented on a screen (e.g., 30) produce higher item nonresponse rates (Mavletova and Couper, 2016a ^a).	The effect of number of items presented on a page on validity and reliability measures.

(Table adapted from Couper et al. 2017:144)

Table 7.3 Measurement error in mobile web surveys: main findings and possible directions for further research.

	Main findings	Possible directions for further research
Grids versus item-by-item or single-item-per-page format	Some advantages of using an item-by-item or single-item-per-page formats in mobile web surveys. A higher correlation in the item-by-item than in the grid format (Lattery et al., 2013). A lower level of straightlining in the item-by-item or single-item-per-page formats (McClain and Crawford, 2013; Peterson et al., 2013^a). Grids in PC web versus single-item per page in PC web versus single-item per page in mobile web: no significant differences between the conditions in item means (de Bruijne and Wijnant, 2013^a).	Dealing with grid questions between mobile and PC web respondents: grids for PC and an item-by-item format for mobile versus item-by-item for both survey modes; effects on validity and reliability.

(Table adapted from Couper et al. 2017:144)

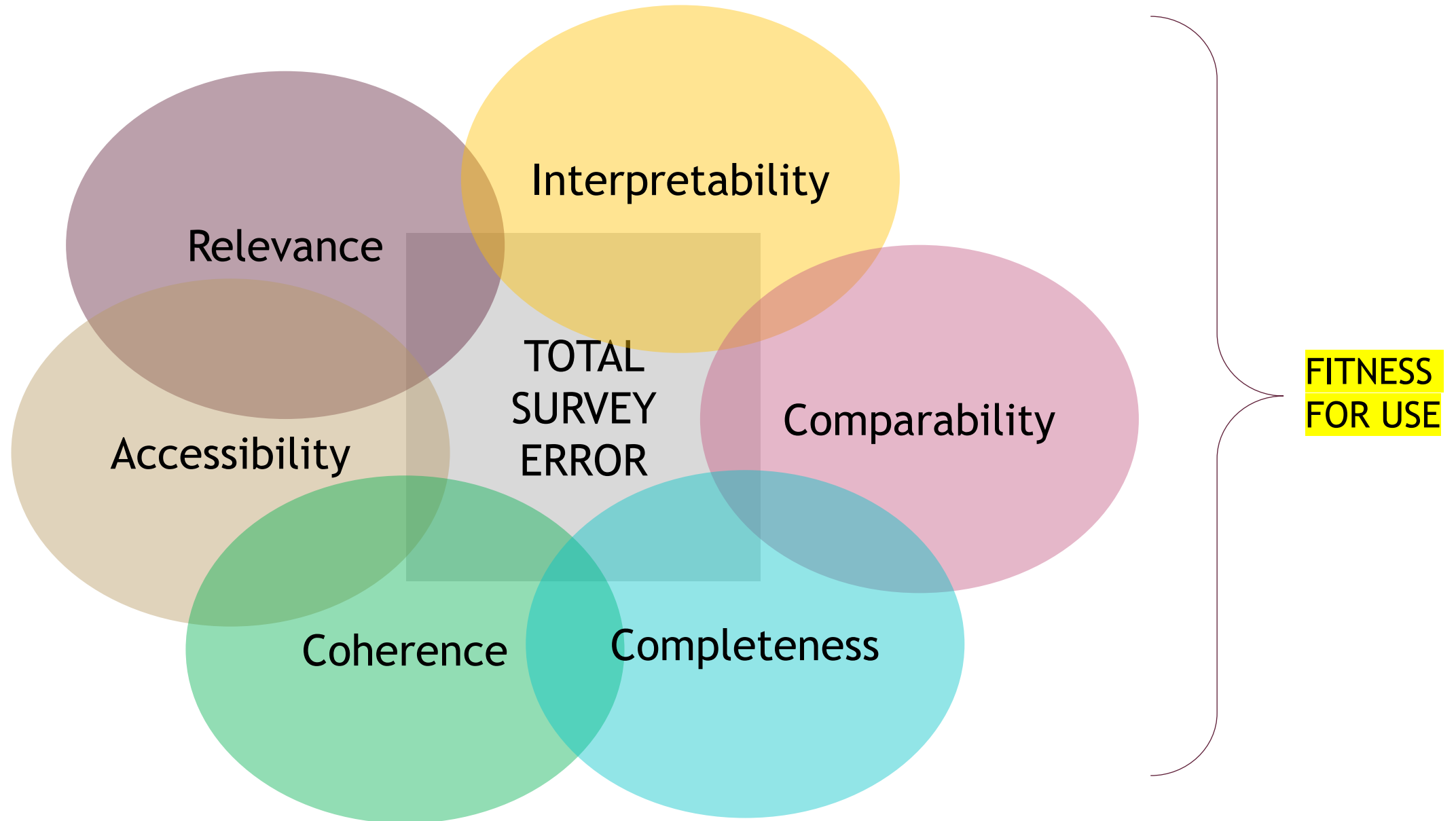
Reducing measurement error

Table 10.4 Estimated odds ratios for logistic regression models predicting top-box or top-two-box score.

Item	Smartphone vs. PC/Legacy					PC/New vs. PC/ Legacy	Smartphone/ New vs. Smartphone/ Legacy
	Legacy	New	Drop- down	Numeric	Slider		
5-point response scale ^{a)} (top-box score)							
The technology to combat global warming is already available; we just need to use it.	0.83	0.79	1.42	1.43	0.84	0.81	1.06
Count of significant differences (total with $p < 0.05$)	4	2	5	4	2	5	0
Significant differences as a percent of total comparisons (17)	24%	12%	29%	24%	12%	29%	0%

(Table adapted from Peterson et al. 2017:215)

From survey **error** to survey **quality**



FROM SURVEY ERROR TO SURVEY QUALITY

Table 1. Common Dimensions of a Survey Quality Framework

Dimension	Description
Accuracy + precision	Total survey error is minimized
Credibility	Data are considered trustworthy by the survey community
Comparability	Demographic, spatial, and temporal comparisons are valid
Usability/Interpretability	Documentation is clear and metadata are well-managed
Relevance	Data satisfy users needs
Accessibility	Access to the data is user friendly
Timeliness/Punctuality	Data deliveries adhere to schedules
Completeness	Data are rich enough to satisfy the analysis objectives without undue burden on respondents
Coherence	Estimates from different sources can be reliably combined

(Table adapted from Biemer 2010:819)

Summary questions

- Main sources of error in the survey process
- What is the total survey error (TSE) framework?
- Difference between random and systematic error
- How can you quantify the total error in survey estimates?
- Survey error vs survey quality

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Other resources

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Thank you

Annibale Cois

Education Building, 4th floor, Room 4060B

acois@sun.ac.za